

Use-Case Flow Specification

Hyperlocal Courier Dispatch & Tracking Engine

Use Case: UC-05 — Complete Delivery

Primary Actor: Delivery Rider | Secondary Actor: Recipient

Preconditions

- The courier request has been created (UC-01) and successfully matched to an active Delivery Rider (UC-02).
- The Delivery Rider has picked up the package and its status is "In Transit."
- The Rider's mobile app has an active network connection to receive/submit OTP verification.

Postconditions

- Success: The order status is updated to "Delivered," a delivery timestamp is recorded, and the Sender Client and Recipient are notified.
- Failure: The order remains "In Transit" or is flagged "Delivery Exception" for support follow-up.

Main Success Scenario

1. Delivery Rider arrives at the drop-off location and selects "Complete Delivery" in the rider app.
2. System triggers use case UC-06: Verify Destination OTP («include»).
3. System sends a one-time password (OTP) via SMS to the Recipient's registered mobile number.
4. Recipient reads the OTP and shares it with the Rider (or enters it directly in a handoff screen).
5. Rider enters the OTP into the app.
6. System validates the OTP against the generated code and checks it has not expired (5-minute window).
7. System marks the order status as "Delivered" and records the delivery timestamp and GPS location.
8. System sends a delivery confirmation notification to the Sender Client and the Recipient.
9. Use case ends successfully.

Alternate Flow — 5a. Incorrect or Expired OTP

- 5a1. System displays an "OTP incorrect or expired" message and does not mark the order as delivered.
- 5a2. Rider selects "Resend OTP"; system generates and sends a new OTP to the Recipient, restarting the 5-minute expiry window (maximum 3 resend attempts).
- 5a3. If OTP verification fails after 3 attempts, the system flags the order as "Delivery Exception," notifies the Sender Client, and routes the case to support for manual resolution.